

Lab: Teaching Through Active Inference

Group Activity — Action

Objective

Practice teaching and learning a crochet technique with deliberate attention to the active inference loops of both teacher and learner, and experience collective decision-making as a group action.

Materials

- Yarn and hooks for all participants
- A list of 4-5 techniques of varying difficulty (prepared in advance)
- Index cards for recording observations
- A timer

Duration

55-65 minutes

Activity Steps

Step 1: Choose Teaching Pairs (5 minutes)

Pair up so that each pair includes someone who knows a technique and someone who does not. (If your circle is small, one teacher can work with two learners.) The technique should be something specific and teachable in 10-15 minutes — examples: - A new stitch (popcorn, bobble, front post double crochet) - A technique (invisible join, magic ring, standing stitch) - A construction method (working in the round, increasing evenly)

Step 2: Teach with Awareness (15 minutes)

The teacher teaches the technique while both teacher and learner pay deliberate attention to the active inference process:

Teacher: Before you start, predict what will be hardest for the learner. As you teach, notice: - What explanation or demonstration did you try first? - Where did the learner struggle? (Their prediction error) - How did you adjust your approach? (Your updated model of the learner) - Did you switch from verbal to visual to physical guidance?

Learner: As you learn, notice: - What did you expect the stitch to feel or look like? - Where did reality differ from your expectation? (Your prediction error) - What helped you most — words, watching, or having your hands guided? - At what point did the stitch "click"?

Both parties jot brief notes on index cards during natural pauses.

Step 3: Debrief the Teaching Loop (10 minutes)

Each pair shares with the group: 1. What technique was taught? 2. What was the teacher's biggest prediction error about the learner? (Where did they misjudge what the learner needed?) 3. What was the learner's biggest prediction error? (Where did the stitch surprise them?) 4. How did the teacher adjust? Did they try multiple strategies? 5. What teaching channel worked best — verbal, visual, or physical?

Step 4: Collective Decision Exercise (15 minutes)

The whole circle now does a group decision activity. Present 3 possible mini-projects that the group could make together (choose from your prepared list, e.g., a set of coasters, a small garland, a color-block scarf).

For each option, discuss as a group: - **Predicted difficulty:** How hard will this be for each member? (Consider the range of skill levels.) - **Predicted satisfaction:** How rewarding will the result be? - **Coordination cost:** How much do individual contributions need to match?

Make the decision together. Notice: - How do different generative models contribute different assessments? - Is there disagreement? How is it resolved? - Does the group converge on a choice that minimizes collective expected free energy (balancing challenge with achievability)?

Step 5: Begin the Collective Project (10 minutes)

Start the chosen project together. Even 10 minutes of working on a shared project lets you experience: - Coordinating individual actions toward a group goal - Checking that your contribution matches the group's shared prediction (same gauge, compatible colors) - The social pleasure of working toward something together

Reflection Prompts

- How was teaching in person different from following written instructions or a video?
- As a teacher, what did you learn about the learner that you did not know before?
- As a learner, what was the most effective thing the teacher did? What would you ask them to do differently?
- During the group decision, did you feel your preferences were heard? How did the group balance individual desires with collective feasibility?

Extension

Continue the collective project across future meetings. At each meeting, briefly check: Is the project matching the group's original prediction? Have there been collective prediction errors (pieces that do not fit, a design choice that is not working)? How does the group adjust? This ongoing project becomes a living laboratory for collective active inference.